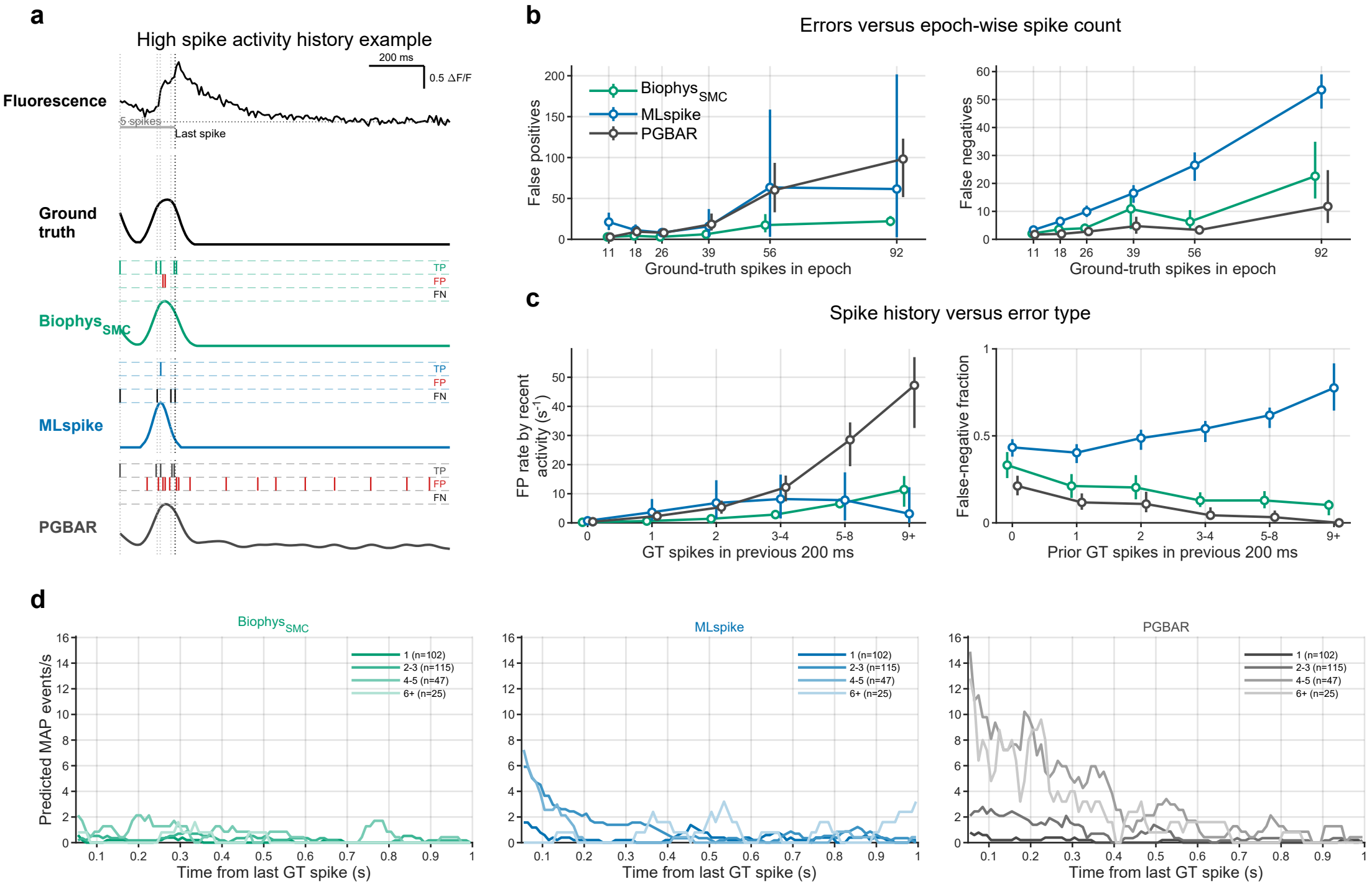

a
High spike activity history example
200 ms
0.5 ΔF/F
Fluorescence
5 spikes
Last spike
Ground truth
TP
FP
FN
Biophys SMC
MLspike
PGBAR
b
Errors versus epoch-wise spike count
False positives
False negatives
Biophys SMC
MLspike
PGBAR
Ground-truth spikes in epoch
c
Spike history versus error type
FP rate by recent activity (s-1)
False-negative fraction
GT spikes in previous 200 ms
Prior GT spikes in previous 200 ms
d
Predicted MAP events/s
1 (n=102)
2-3 (n=115)
4-5 (n=47)
6+ (n=25)
Time from last GT spike (s)